

The Beat-SPY Challenge

Process, findings, and exactly how the stocks were chosen

Watchtower / Claude — build day 2026-09-07 (Labor Day) — prepared for Eric Konoski

One-paragraph summary. The task was to build a trading system that beats the S&P 500 (SPY, dividends reinvested) on a sealed window it had never seen, under rules written down before the window was opened: a higher compound return AND a drawdown no deeper than SPY's. Fifty-five variants across nine generations were graded on 2006–2023. The survivor, Compass v7, holds one equity-index ETF at a time by 12-month momentum with a 20% washout sleeve in single stocks. On the 18-year build window it did 11.96%/yr with a –30% worst drawdown against SPY's 9.61%/–55%, and beat in both halves of the window. On the sealed window (2024-01-02 → 2026-09-04) it returned 24.4%/yr against SPY's 21.4% but drew down –22.1% against –18.8%, so it FAILED the rule as written. Single-stock selection, with and without real point-in-time fundamentals, never beat the index book at equal risk. The most valuable findings of the day were three data defects that had manufactured the first 'winning' result.

1. The rules, frozen before any number

- Benchmark: SPY total return, dividends reinvested at the ex-date close, from \$100,000.
- Beat = (1) higher compound return AND (2) maximum drawdown no deeper than SPY's, on end-of-day equity. A higher return with a deeper drawdown is leverage, not edge, and does not count.
- Instruments: US stocks and ETFs; options only with defined risk (long calls/puts, debit spreads); total option premium at risk never above 25% of equity; no margin; no leveraged/inverse ETFs.
- Fills only at prices that printed: the close of the signal bar or a later bar. Shares cost 5 bps per side. Modeled option prices carry a 2% spread charge per side plus \$0.65 per contract.
- No lookahead. Survivorship must be stated. Build window = all data through 2023-12-31, inspected freely. Sealed window = 2024-01-02 through 2026-09-04, run ONCE on frozen written rules. Any change after seeing sealed results forfeits the window.
- Tie-break if both beat: return per unit of drawdown. A failed holdout is reported in the same detail as a pass.

2. The measuring instrument

Everything ran through one simulator (analysis/beat_spy.py) over Watchtower's own daily price table (Polygon, split-adjusted), marking equity every trading day. The ETF pool was backfilled to 2005 from the vendor where our history started late. SPY's dividends were fetched and cached so the benchmark is a true total return. Each variant is a named parameter set; every run stores its parameters, daily equity vs SPY, and full trade list, so any number in this document can be re-derived from the database. A sealed run refuses to execute twice for the same name.

3. The build sequence, generation by generation

All returns are compound annual growth rates on 2006-01-03 → 2023-12-31 unless stated. SPY total return over that window: 9.61%/yr, worst drawdown –55.2%. Numbers marked (pre-guard) were later found to contain phantom trades — see section 4 — and were re-run; the clean number follows.

v1 — the first idea (pre-guard 4.66%, clean 3.69% / –39%)

Forty-seven ETFs (indexes, SPDR sectors, industries, themes, TLT, GLD), top 5 by 6-month momentum, weekly re-ranking, liquidated whenever SPY closed under its 200-day; plus a 20% sleeve buying deep 16-day green-dot bottoms on stocks 50%+ off their highs with a three-tranche ladder. Verdict: the forced liquidation was a whipsaw

machine (172 exits at 31% win, −\$64k), and the thematic funds let the ranker buy the 2020–21 bubble cohort at the top.

v2 — Faber monthly regime, no forced selling (6.70–7.65%)

Regime checked against SPY's 10-month average on month-end closes only; holdings leave on their own merits. Lesson: with no regime at all the drawdown went to −48%; the regime is worth ~15 points of drawdown and costs nothing in return.

v3 — core pool, dual momentum, first option overlay (6.1–6.7% / −25 to −30%)

Thematic funds removed ex ante; a fund must also be UP over its own lookback (Antonacci's absolute momentum). A deep-in-the-money call overlay (0.80 delta, 9-month tenor, Black-Scholes on realized vol × 1.15, rolled at 60 days) LOST return: rotations expressed as calls bled −\$160k at 34% win while long holds made +\$272k. Calls pay on long trends and bleed on churn.

v4 — index-only concentration (4.3–6.4%)

One index at a time. The trade list exposed the real problems: weekly re-ranking among correlated indexes flipped every 7–14 days; TLT and GLD competed with stocks in risk-on; and the 200-day re-entry sat in TLT from March to August 2020 while QQQ ran off the low, and in TLT through all of 2009.

v5 — Global Equities Momentum as published (clean 7.10%; skip-21 variant 9.76%)

Monthly rebalance, equity indexes only, absolute momentum as the single filter, bonds only when no index qualifies. The 12-1 variant (12-month return skipping the most recent month) was the first cell to beat SPY on both clauses — by 0.15 points, before the leveraged-ETP cleanup.

v6 — fast re-entry and vol targeting (refused)

A two-speed regime (leave on the 200-day, re-enter on a rising 50-day) took 65 forced exits for −\$54k: 1.8–5.8%. Vol targeting was flat without leverage and worth half a point with 2× calls. Bonds-only defensive (dropping gold, which v5 had held through its 2008 crash) added +1.2 points and was kept.

v7 — the candidate (11.96% / −30.3%)

Skip-21 plus bonds-only. Robustness: skips of 15/21/30/42 all beat SPY (a plateau, not a lucky cell); lookbacks of 189 and 315 trail SPY slightly, so 252 is a published prior we lean on. It beats in BOTH halves: 2006–2015 7.99% vs 6.96% (drawdown −23% vs −55%); 2016–2023 17.26% vs 13.15% (−30% vs −34%). The index sleeve alone, with no single stocks at all, beats on drawdown in both halves and ties SPY on return in the modern half; the dot sleeve supplies the modern-era outperformance.

v8 — 'what about great returns?' (single names refused; 3× calls 14.14%)

Single-name 12-1 momentum on a universe that was liquid AT EACH DATE (≥\$10M/day, ≥\$5): top 5/10/20 graded 6.8–10.4% at drawdowns of −58% to −80%; 2008 −56%, five straight red years 2014–18, 2019 +104%. A lottery profile. Longer dot holds traded half a point of return for three of drawdown. The 3× call overlay printed 14.14% / −30.3% and beat in both halves — but see section 7.

v9 — point-in-time fundamentals on single names (11.34% / −48%; still refused)

Real filing dates exist for 2,100–4,000 names a year back to 2008. A quality gate (four fresh quarters, positive trailing earnings, operating cash flow above earnings) lifted the stock book from 6.8%/−74% to 11.3%/−48% — the largest single-rule improvement of the day, and 2008 went from −56% to −5%. Ranking BY revenue growth instead of momentum collapsed to 5.8%. It still failed the second half on both clauses (12.9% vs 13.2%; −48% vs −34%,

trough 2020-03-18) and never beat v7.

4. The findings that mattered most: three ways the data lied

- **Ticker reuse (splices).** The first 'beat' (10.69%) rested on one trade: symbol AI bought at \$2.83 in July 2020 and sold at \$134.54 for +\$118k. That symbol was Arlington Asset until C3.ai took it in December 2020; the price table ran the two companies as one series. TNL (2010→2021) was the same shape.
- **Multi-year holes.** TFIN had 2008 bars, then nothing until 2022. A bar-count exit stalled across the gap, so a ladder 'held' 14 years for a phantom +\$52k, present in 28 of the stored runs.
- **Leveraged and inverse ETPs.** About 30 daily-reset products (JNUG, NUGT, SCO, SQQQ, TSLQ...) sat in the 'stocks in drawdown' universe. Removing them RAISED the dot sleeve (7.10% → 8.40%): they were net losers.
- **The fixes, now permanent.** A close 3× (or ■) its prior close or a gap over 10 calendar days marks a series defect; a held position exits at the LAST REAL PRINT when its tape breaks; a dot on a tape broken inside two years is refused; every run logs both counts; leveraged ETPs are refused by name; old runs are kept, labeled, and re-run. Census: of 1,538 deep-dot tickers, 232 carry holes and 101 carry splices.
- **Checked and kept.** FCEL's 2019 six-bagger: the stored price levels are off by a split factor, but every daily ratio is a real print, and P&L is ratio-based.
- **Effect on the record.** The phantoms were worth about one point of annual return in every dots-carrying variant. All 55 variants were re-run clean before anything was frozen.

5. Exactly how the stocks were chosen

5a. The index sleeve (80% of equity) — this is what the frozen system trades

- Universe: SPY, QQQ, IWM, MDY, DIA, RSP. Defensive: TLT only.
- On the last trading day of each month, for every fund: momentum = close 21 trading days ago ÷ close 252 trading days ago – 1. Twelve months, skipping the most recent month.
- A fund qualifies only if its momentum is positive. Rank the qualifying equity funds; hold the top one. A current holding stays while it remains in the top two; otherwise rotate to the best.
- If no equity fund qualifies, hold TLT if TLT's own momentum is positive, else cash.
- Execute at that month-end close, 5 bps per side. No stop inside the month.
- Not used, all tested and refused: SPY-vs-200-day regime, per-fund moving-average filter, vol targeting, fast re-entry, weekly cadence, gold as a defensive asset.

5b. The washout sleeve (20% of equity, 10 slots) — the only single-stock selection in the system

- Universe: stocks with ≥\$10M/day average dollar volume over the trailing 90 days as of the run date (a 'liquid today' universe — survivorship stated), price ≥\$2, leveraged/inverse ETPs excluded by name, any ticker with a series defect in the prior two years refused.
- Signal: a 16-trading-day wavetrend cross-up (fixed-anchor blocks on SPY's calendar) with cross depth ≤ –30, on a stock ≥50% below its 2-year high. This is Eric's green-dot read, graded on 20,404 dots in August: deep dots on deep drawdowns pay a median +20% over a year, 57% positive; shallow crosses do not.
- Entry: three equal tranches — the first at the dot bar's close, then resting limits 15% and 25% below it, filled only when a later bar's low prints through.
- Exit: everything sold at the close 126 trading days after the dot. No stop, no target. If the tape breaks first, sold at the last real print.

- Nothing about the company is read: no earnings, no revenue, no sector. It is purely price structure, and it carried the 2016–2023 outperformance.

5c. Single-name momentum (v8) — built, graded, refused

- A ranking table with one row per SPY month-end per ticker that was liquid AT THAT DATE (60-bar average dollar volume $\geq$ \$10M, close $\geq$ \$5), holding the 21- and 252-day-ago closes. 158,217 rows over 218 months; about 725 names a month.
- Rank by 12-1 momentum, keep positive momentum only, hold the top N equal-weight, monthly. A name whose tape breaks mid-hold leaves at the last real print. A thin month puts one slot in TLT.
- Result: 6.8–10.4%/yr at –58% to –80% drawdowns. Refused on drawdown in every configuration.

5d. Fundamentals gates (v9) — built, graded, refused as a book, kept as a veto

- For each (month-end, ticker): trailing-four-quarter net income, operating cash flow, and revenue, plus the prior four quarters' revenue, built only from filings whose report date is on or before the month-end, taking the EARLIEST filing per period so a later restatement can never leak back. 115,262 rows.
- Quality gate: four fresh quarters (last filing within 130 days), trailing net income > 0 , trailing operating cash flow > 0 and above net income. Growth gate: eight quarters, trailing revenue up more than 10%.
- Survivors ranked by momentum (or, in one variant, by revenue growth). Top 5/10/20, with and without the SPY 200-day regime.
- Result: quality + momentum 11.34% / –48.3%; quality + growth 10.70% / –46.8%; growth-ranked 5.82%. The gate is worth ~26 points of drawdown on single names and is kept as a proven veto. It does not produce a book that beats holding the right index.

6. The frozen system and its build record

2006-01-03 → 2023-12-31	CAGR	Max DD	2006–2015	2016–2023
SPY total return	9.61%	–55.2%	6.96% / –55.2%	13.15% / –33.7%
Compass v7 (primary, shares)	11.96%	–30.3%	7.99% / –23.4%	17.26% / –30.3%
v8_lev3 (declared options expression)	14.14%	–30.3%	9.49% / –23.4%	20.38% / –30.3%
v7 index sleeve alone (no stocks)	11.19%	–28.6%	9.76% / –27.9%	13.17% / –28.6%

Sleeve split on the build window: index sleeve 29 trades, 21 wins, +\$463k; dot sleeve 273 trades, 176 wins (64%), +\$199k, average win \$1,850 / average loss –\$1,305. Frozen in writing (docs/research/compass_frozen_rules.md) at commit c039634; the sealed run shipped at commit 153dff1.

7. The sealed window: run once, failed

2024-01-02 → 2026-09-04	CAGR	Max DD	\$100k became	Trades
SPY total return	21.39%	–18.8%	\$167,862	—
Compass v7 (primary)	24.43%	–22.1%	\$179,313	63 (37 wins)
v8_lev3 (options)	24.43%	–22.1%	\$179,313	identical — zero calls opened

- Return clause: passed by 3.0 points. Drawdown clause: FAILED by 3.3 points. Both troughs 2025-04-08, the tariff crash, with the book in QQQ. Tie-break (return per unit of drawdown) also to SPY, 1.11 vs 1.14.
- Year by year: 2024 +24.6% vs +25.6%; 2025 +18.5% vs +18.0%; 2026 to date +20.3% vs +13.3%.

- The entire index sleeve was three trades: QQQ from 2024-01-02, SPY from 2025-04-30, QQQ from 2026-02-27. Dots: 60 trades, 34 wins, +\$20.4k.
- Why it failed: the system's edge is sitting out bear markets (2008, 2022). A 32-month window with no bear gave the absolute filter nothing to do and left only the cost of concentration. The stated odds beforehand were exactly backwards.
- **The options expression was hollow.** In the sealed window the 25% premium cap refused every 3× call, so the run collapsed to shares. On the build window its 14.14% rested on FIVE call trades in 18 years, two of them QQQ 2017–18 worth \$161k. New rule: any overlay states how many positions its number rests on.
- The record stands as run. A new idea is a new name on a new window, never a second look at this one.

8. Everything that was refused, and why

Idea	Best result	Why refused
Forced liquidation on SPY < 200-day	3.7% / –39%	172 forced exits at 31% win; whipsaw
Thematic ETFs in the pool	—	ranker buys the bubble cohort at the top
Weekly re-ranking of indexes	6.0% / –32%	7–14 day flips among correlated funds
Gold as a defensive asset	–\$19.6k in 2008	momentum held GLD through its crash
Two-speed regime (50-day re-entry)	5.8% / –32%	65 forced exits, –\$54k
Vol targeting	9.2% / –30% (2× calls)	flat without leverage, half a point with
Call overlay on a churning pool (v3)	6.5% / –30%	–\$160k on rotations at 34% win
Single-name momentum (v8)	10.4% / –80%	lottery profile; 2008 –56%; five red years
Longer dot holds (252 days)	11.5% / –27%	–0.5 pt return for +3 pts drawdown
Heavier dot sleeve (40–50%)	12.5% / –32%	failed the first half of the window
Fundamentals-gated stock book (v9)	11.3% / –48%	fails 2016–23 on both clauses; never beats v7
Ranking by revenue growth	5.8% / –54%	fundamentals work as a veto, not a picker
3× call overlay as the frozen expression	14.1% / –30%	five trades in 18 years; zero out of sample

9. The checks a result had to pass

- **Neighbors agree.** A beat that lives in one parameter cell is a curve fit. Skip 15/21/30/42 all beat; that is a plateau. Lookback 189/315 trail; stated as a prior, not a plateau.
- **Both halves.** Every headline number split at 2015-12-31 and required to beat on both clauses in each half. v7 passes; v8's stock book and v9's fundamentals book each fail the modern half on drawdown.
- **The sleeve alone.** The index sleeve with no single stocks beats on drawdown in both halves, so the edge does not depend on the survivorship-tainted part.
- **Every trade inspected.** The trade lists, not the summary numbers, found the AI splice, the TFIN hole, the gold-in-2008 problem, the 2020 re-entry lag, the five-trade options overlay.
- **Holes are holes.** A missing bar, a missing filing, an exit after the last bar — each is stated, never counted as zero.

10. Conclusions and the honest ceiling

- A fully automated, long-only system that must hold a drawdown no deeper than SPY's lands at roughly 12%/yr in shares on this data, ~14% with defined-risk calls, at a drawdown around –30% against SPY's –55%. That is the ceiling of this frame, not of trading.

- Picking stocks — by momentum, by quality, by growth — did not add return over holding the right index at equal risk. It added risk. An index already owns every winner and sheds every loser for free; single-name selection has to buy that diversification back with screens, and the screens cost more than they earned.
- Fundamentals are real: the quality gate is the largest single-rule improvement measured all day. They earn a veto worth ~26 points of drawdown on single names, not a way to beat the index.
- The one single-stock edge that graded positive is the 16-day deep green dot on stocks in major drawdown — price structure, not company quality — and it is already in the frozen system at 20%.
- The system's advantage is a bear-market advantage. Over any stretch that includes one it should beat; over a stretch that does not, it will trail on drawdown by the cost of concentration, as it just did.
- The marketable asset is the measured process and the clean record, including this failure, reported in the same detail as a pass.

What happens next

- The journal now grades Eric's own account by the same two-clause rule against SPY total return (watchtower_journal, scoreboard section), so the human is measured the way the machine was.
- The green-dot record is being recounted under the series-defect filter and under Market Cipher's own wave settings (9/12/3), tagging the 'bottom buy' dot beside the current definition.
- No further sweeps on the build window: that is how the first false 'win' was produced.

Appendix — the clean grid (hygiene v2, build window, SPY 9.61% / –55.2%)

Variant	CAGR	Max DD	Beats?
compass_v1	3.69%	–39.2%	no
v1_top3	5.33%	–40.5%	no
v1_top8	5.06%	–29.2%	no
v1_mom63	2.90%	–38.9%	no
v1_mom252	6.31%	–35.1%	no
v1_no_dots	2.55%	–36.9%	no
v1_a90	3.46%	–44.6%	no
v1_dots_only	2.48%	–18.3%	no
v2_faber_liq	6.13%	–42.0%	no
v2_noregime	3.66%	–46.8%	no
compass_v3	6.06%	–30.3%	no
v3_top3	6.08%	–32.3%	no
v3_top8	5.60%	–26.3%	no
v3_no_dots	5.40%	–32.5%	no
v4_index_top1	4.97%	–30.8%	no
v4_index_top2	4.59%	–24.7%	no
v4_index_top1_nodots	4.26%	–37.9%	no
v4_index_top1_dots40	5.38%	–28.7%	no
v4_dots_heavy	4.89%	–31.5%	no
v4_index_top1_calls_1p5	4.58%	–32.3%	no
v4_index_mom126	6.39%	–26.8%	no
v5_gem	8.40%	–36.1%	no
v5_gem_nodots	6.79%	–34.1%	no
v5_gem_skip21	11.39%	–30.3%	yes
v5_gem_top2	9.48%	–34.8%	no
v5_gem_faber	5.83%	–36.1%	no
v5_gem_calls_1p5	9.44%	–32.0%	no
v5_gem_dots40	6.86%	–37.6%	no
v5_core_monthly	7.47%	–35.9%	no
v6_ts_monthly	4.89%	–31.3%	no
v6_ts_weekly	4.44%	–36.8%	no
v6_ts_mom126_weekly	7.28%	–32.9%	no
v6_vt15	7.78%	–35.5%	no
v6_vt15_lev1p5	7.43%	–35.4%	no
v6_vt20_lev2	9.24%	–29.9%	no
v6_ts_vt15_lev1p5	3.78%	–31.0%	no
v6_ts_vt15_lev1p5_weekly	3.32%	–35.6%	no
v6_ts_vt15_lev1p5_nodots	2.10%	–31.0%	no
v6_ts_bonds	7.16%	–30.4%	no

Variant	CAGR	Max DD	Beats?
v6_gem_bonds	9.60%	−36.1%	no
v6_ts_vt15_lev1p5_bonds	5.47%	−27.1%	no
v7_skip21_bonds (FROZEN primary)	11.96%	−30.3%	yes, both halves
v7_skip10	9.56%	−36.1%	no (by 0.05)
v7_skip15	10.38%	−29.6%	yes
v7_skip30	9.98%	−29.7%	yes
v7_skip42	11.61%	−29.4%	yes
v7_mom189_skip21	8.92%	−29.9%	no
v7_mom315_skip21	9.41%	−29.0%	no
v7_skip21_bonds_top2	10.40%	−33.1%	yes
v7_skip21_bonds_nodots	11.19%	−28.6%	yes, both halves
v7_skip21_bonds_vt20_lev2	12.23%	−26.8%	yes
v7_skip21_bonds_calls_1p5	12.04%	−27.0%	yes
v8_stk5	10.39%	−79.5%	no
v8_stk10	6.80%	−74.0%	no
v8_stk20	8.32%	−70.0%	no
v8_stk10_reg200	10.39%	−57.8%	no
v8_stk10_nodots	6.00%	−75.4%	no
v8_stk10_dv50	7.90%	−72.5%	no
v8_dots252	11.46%	−26.9%	yes
v8_dots252_single	11.78%	−34.4%	yes
v8_dots252_heavy	12.49%	−32.0%	fails H1
v8_lev2	13.52%	−29.8%	yes
v8_lev3 (FROZEN options expression)	14.14%	−30.3%	yes; 5 call trades
v9_q10	11.34%	−48.3%	fails H2
v9_g10	11.79%	−59.5%	no
v9_qg10	10.70%	−46.8%	fails H2
v9_qg5	10.46%	−54.4%	no
v9_qg20	10.32%	−47.7%	no
v9_qg10_reg200	8.78%	−45.0%	no
v9_growthrank10	5.82%	−53.7%	no
v9_q10_nodots	11.07%	−54.3%	no

'Beats?' is the build-window rule (higher CAGR than SPY and drawdown no deeper than −55.2%); the both-halves check is the stricter bar applied before anything was frozen. Every row can be re-derived from `beat_spy_runs` / `beat_spy_trades` in the Watchtower database.